

GETTING STARTED

Join a game in minutes

For the Player

- 1 Open [grammarx.onrender.com](#) in your browser
- 2 Enter your player name and press **Join**
- 3 Select the **Player** role on the welcome screen
- 4 Enter the **6-character room code** shared by the host
- 5 Wait in the lobby until the host starts the game
- 6 Answer fast and correctly — points depend on speed!

For the Teacher / Host

- 1 Enter your name and select the **Host** role
- 2 A **room code** and QR code are generated automatically
- 3 Display or share the QR code — players scan it with their phone
- 4 Choose the **game mode** and time per round (10s – 60s)
- 5 Press **▶ START** and advance round by round from the dashboard

COMPATIBLE WITH

✓ Chrome · Firefox · Safari · Edge

✓ PC / Mac · Tablet · Smartphone

✓ No install needed — just a browser



ACCESS RIGHT NOW
grammarx
.onrender.com

GRAMMARX V1.0 · 2026
ENGLISH GRAMMAR EDUCATIONAL PLATFORM

A multiplayer web application that gamifies English grammar learning. Designed for classroom use, it lets teachers and students practise four key grammar structures through competitive real-time matches.

Project Information

Subject	English 2
Institution	Universidad Privada Domingo Savio
Lecturer	Ruth Gascher
Team	Jonathan Campos Luis Rodrigo Vargas Melgar Carla Noelia Cano Diego Justiniano Viera
Year	2026
Deploy	Render Cloud ↗

Technologies used

- REACT + VITE
- NODE.JS
- SOCKET.IO
- EXPRESS
- SQLITE
- RENDER

Project objectives

- ◆ Build an engaging and fully functional educational tool
- ◆ Apply modern full-stack technologies (React + Node.js)
- ◆ Demonstrate bidirectional real-time communication
- ◆ Deploy a complete SPA to a production environment

"Technology doesn't replace the teacher — it gives them superpowers."

GRAMMARX

EDUCATIONAL PLATFORM · ENGLISH GRAMMAR

Master English grammar while competing in real time

Kahoot-style · Multiplayer · 4 modes · 5 levels

**Time Machine**
PAST SIMPLE

**Duel Mode**
COMPARATIVES

**Evidence File**
PRESENT PERFECT

**Crystal Ball**
FUTURE TENSE

5 DIFFICULTY LEVELS

Beginner

Easy

Medium

Hard

Expert

- MULTIPLAYER
- REAL TIME
- NO SIGN-UP
- OPEN SOURCE

4
GAME MODES

40+
QUESTIONS PER MODE

5
DIFFICULTY LEVELS

∞
UNIQUE SESSIONS

How does it work?

- 1 Host creates a room — a 6-character code and QR are generated
- 2 Players join from their phone or PC, no installation needed
- 3 Host picks the game mode and time per round (10s – 60s)
- 4 10 rounds of random questions — fastest and most accurate player wins



SCAN TO PLAY NOW
grammarx
.onrender.com
No sign-up · Just a browser

THE PROJECT

Context and motivation

The Problem

English grammar is often taught through repetitive, static exercises that generate low motivation in students. The lack of immediate feedback and the absence of game-like elements limit the retention of key concepts.

The Solution: Gamification

GRAMMARX transforms grammar practice into a competitive and collaborative real-time experience, where speed and accuracy are both rewarded.

- Instant feedback per answer (correct / wrong)
- Configurable time pressure that boosts concentration
- Friendly competition between peers in the same classroom
- 100% random questions — every game is unique

Target audience

- English students at A2–B2 level (CEFR)
- English teachers in in-person or hybrid environments
- Continuing education courses and self-study learners

Grammar content

- Past Simple** — regular & irregular verbs, sentence order
- Comparatives & Superlatives** — formation rules, exceptions
- Present Perfect** — have/has + past participle, reading comprehension
- Future Tense** — will vs. going to, predictions and plans

GAME MODES

GRAMMARX

The 4 game modes



Time Machine

PAST SIMPLE · SYNTAX

The words of a Past Simple sentence are shuffled. The player must tap the words in the correct order to reconstruct the sentence before time runs out.

- Regular verbs
- Irregular verbs
- Word order



Duel Mode

COMPARATIVES & SUPERLATIVES

Four answer choices appear on screen. The player selects the correct comparative or superlative form of the given adjective, applying the rules for short and long adjectives.

- ADJ+er than
- the ADJ+est
- more / most ADJ



Evidence File

PRESENT PERFECT · COMPREHENSION

A short professional-context passage is shown. The player reads it and answers a comprehension question related to completed actions using the Present Perfect.

- have / has + PP
- Comprehension
- Work context



Crystal Ball

FUTURE TENSE · SELECTION

An incomplete sentence with a blank is presented. The player selects the correct future form from four options: will, going to, won't, or other variants.

- will / won't
- going to
- Predictions

TECH STACK

Architecture and technology



Frontend

React 18 + Vite 5	Single-file SPA
Socket.io Client	Real-time WebSocket
Bricolage Grotesque	Display typeface
JetBrains Mono	Monospace typeface

Backend

Node.js + Express	HTTP + WS server
Socket.io Server	Bidirectional events
SQLite (node:sqlite)	Native persistence
Render	Production deploy

Technical features

- Randomly generated 6-character room code per session
- Auto-rendered QR code via QRCode.js library
- 2 random questions per difficulty level selected (10 rounds)
- Timer synchronised between host and players via Socket
- Full dark palette: background #080810, gold accent #f0c040
- Responsive design: host on desktop, players on mobile

40+

QUESTIONS
PER GAME

5

DIFFICULTY
LEVELS

10

ROUNDS PER
MATCH

∞

UNIQUE
SESSIONS

"The combination of Socket.io and React enables real-time synchronisation that makes every match dynamic and smooth, with no reloads or noticeable latency."